# Agility & Leadership Improvement Plan: Once-Only Technical System (OOTS)

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## 1 Diagnostic

### 1.1 Business Context

I work as an independent contractor for DIGIT-EU, the Directorate-General for Digital Services at the European Commission. My work is on the Once-Only Technical System (OOTS), a pan-European infrastructure that lets public authorities exchange documents securely across borders. Citizens and businesses then do not have to submit the same information repeatedly [1].

OOTS retrieves data directly from the official source, with the user's explicit consent. It rests on four pillars: a **decentralized architecture** connecting existing national systems; **interoperability enablers** such as eID and eDelivery [2]; **user control and consent**; and **standardized evidence** [3].

OOTS is not a commercial enterprise and does not compete in a market. Its mission is to promote cooperation among sovereign nations and serve the public interest.

**Environment (VUCA/BANI).** The environment is volatile and complex. Political and legal factors dominate: twenty-seven Member States with different governments and timelines, plus the GDPR, eIDAS, the Once-Only Regulation and national laws. Each exchange must satisfy both Member States involved. The factors are interdependent. A political decision to strengthen data protection creates legal requirements, which demand technological solutions that citizens must accept. This is the non-linear dynamic that BANI describes. Cybersecurity threats add brittleness and anxiety to a system handling sensitive data.

**The agility need.** Agility here does not mean market speed. It means:

- adapting to regulatory change;
- absorbing the feedback of many sovereign stakeholders;
- releasing reliably in iterations (RFCs and TDDs [4]);
- learning quickly when assumptions prove wrong.

The need is high, because the direction is largely implicit and decisions are often reactive (see the strategic reflection). The environment will not become simpler. Reusability for initiatives such as the EU Digital Wallet will widen the scope further.

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## 1.2 Hard Dimensions

### 1.2.1 Strategy

OOTS has no formal strategic statement, but its documentation reveals a clear direction [5]. In my reflection paper I formulated one:

- **Mission:** seamless, secure, consent-driven cross-border data exchange, so nobody provides the same information twice.
- **Vision:** a Union where digital interoperability is the norm and public services work as one system regardless of borders.
- **Values:** decentralization, trust and security, user empowerment, interoperability, reusability and cooperation.

*Readiness assessment.* The purpose is strong and gives a stable backbone. The weaknesses are:

- there are no measurable outcomes, for example procedures supported or Member States connected by a date;
- there is no periodic strategic review. Stakeholder analysis are not formally used, so the strategy is implicit rather than guided;
- some stakeholders, notably data protection authorities and citizens, are not systematically engaged, although they strongly influence legitimacy and trust.

Strategy is collaborative and iterative, which supports adaptability. It lacks the explicit goals and short feedback cadence that would make it agile.

### 1.2.2 Structure

My project team is organized as follows (Table 1).

The team has about 20 people in six functional groups. Each group has a lead, and a single Tech Lead sits at the top.

My role is Application Architect and Backend Team Lead. The business rules of the project are dense and sensitive, so the backend is its most complex and technical part. I have four responsibilities:

- advising the Solutions Architect on technical decisions;
- leading the backend team and securing delivery within milestones;
- implementing the core features and libraries;
- reviewing all backend code.

*Readiness assessment.*

| Team | Composition |
|---|---|
| Common Service Leader | 1 Solutions Architect / Tech Lead |
| Business Analyst Team | 1 BA Lead, 1 Senior BA Analyst |
| | |
| Infrastructure: | |
| DevSecOps Team | 1 DevSecOps Lead, 2 DevSecOps Experts |
| | 1 Vulnerability/Incident Analyst, 1 Risk/Compliance Expert |
| Development: | |
| Backend Team | 1 Application Architect / BE Team Lead (me) |
| | 2 Senior BE Developers |
| Frontend Team | 1 Tech Business Analyst / FE Team Lead |
| | 2 Senior FE Developers |
| | |
| Quality Assurance: | |
| Quality Assurance Team | 1 QA Team Lead |
| | 2 Senior QA Developers |

Table 1: Structure of the project team

- **Strengths:** small, specialised, cross-functional teams, with security, compliance and QA embedded in the project rather than external gates. Roles are clear. The team is not buried in a bureaucratic hierarchy, in contrast with the wider Commission.
- **Weaknesses:** there is a hierarchy of leads. Decisions flow through the Tech Lead, with my advisory input. Reviewing all backend code makes me a potential bottleneck and a single point of failure. Specialisation by function (BA, BE, FE, QA) risks hand-offs instead of fully cross-functional ownership. Larger structures around the project, such as Member States and the Commission, are outside the team's control.

### 1.2.3 Systems

*Readiness assessment.* The toolchain is modern and supports agile practice: version control, automated pipelines and cloud deployment enable frequent, reliable releases. Argo CD gives continuous visibility of what is deployed against what is declared in Git, and Prometheus and Grafana make the runtime behaviour of the system observable through metrics, dashboards and alerts. Together they shorten the feedback loop between a release and its effect in production. The code is open source, which adds transparency [6].

| Purpose | Tool |
|---|---|
| Communication | Microsoft Teams and Outlook |
| Documentation | Confluence |
| Project management | Jira |
| Version control | GitLab |
| Automation | CI/CD pipelines |
| Deployment | AWS / Azure |
| Deployment monitoring | Argo CD (GitOps continuous delivery) |
| Observability | Prometheus (metrics) and Grafana (dashboards and alerting) |

Table 2: Project systems

Weaknesses to probe:

- email remains a communication channel in a distributed team;
- documentation in Confluence can drift from the code;
- the dashboards cover system health, but it is not clear that delivery metrics, such as DORA metrics, are tracked to show how fast the organization learns.

#### 1.2.4 Talent Practices

**Recruitment.** DG DIGIT uses a hybrid model with three contractual frameworks [7], [8]:

- **Permanent staff** are recruited through EPSO. Candidates must pass strict reasoning, situational and technical competency assessments to reach AD or AST grades.
- **Temporary staff** are recruited through CAST. IT specialists register in a permanent pool, and are shortlisted for interviews when OOTS needs specific expertise. Contracts run from 1 to 6 years.
- **External consultants** come through IT framework contracts (outsourcing). Private consulting firms win EU tenders and allocate their own engineers and Scrum masters to the project. Contracts run from 1 to 4 years. Much of the development, testing and maintenance is delivered this way, and I am part of this group.

*Assessment.* The hybrid model gives access to scarce skills and flexible capacity. It also brings risks:

- fixed-term contracts mean knowledge walks out the door;
- the team mixes people with different employers, incentives and loyalties;
- selection through EPSO and tenders measures formal competence but not agility traits such as learning and collaboration.

**Retention.** From my personal experience, four factors keep people:

1. **Compensation.** Salaries are very high. In my view this is the main reason people stay.
2. **Project culture.** Unlike the bureaucratic nature of the EC, the team uses modern development approaches. People focus on designing solutions rather than on dead-end meetings, and they are encouraged to suggest better practices and new ways of working.
3. **Sense of purpose.** People feel they contribute to something bigger than themselves.
4. **Supportive team and recognition.** People help each other and share knowledge. Good consultants are appreciated and can stay longer.

*Assessment.* Retention is strong and partly intrinsic (purpose, autonomy, community), not only financial. Two caveats:

- the pay effect could be fragile if the market changes;
- retention is tied to recognition of individual consultants, and it is not clear that development and career paths are structured.

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## 1.3 Soft Dimensions

### 1.3.1 Culture

Culture here has two layers: the Commission's bureaucratic environment and the project's own modern, engineering-driven culture. The project culture values innovation, focus on the work, knowledge-sharing and cooperation. These are consistent with the values of the strategic statement: cooperation, trust and security, and reusability. Decentralization and respect for sovereignty also shape culture, with consensus-building (RFCs) preferred to imposed decisions. The culture is collaborative and open (open source). Its risk is that consensus can slow decisions, and the surrounding institution pulls back towards bureaucracy.

### 1.3.2 Trust

Trust is a precondition for delegation, fast decisions and experimentation. It differs from psychological safety: trust is about relying on others' competence and intent, while psychological safety is about being safe to speak up [9]. I assess it through five aspects.

1. **Trust in the product.** Trust and security are core values: every exchange is encrypted, authenticated and auditable. Member States connect only if they trust OOTS with sovereign data, and open source and the RFC process build that trust through transparency.
2. **Peer trust.** People help each other all the time. When someone struggles, any member of any team offers help and pairs or meets to solve the problem. For example, QA members meet BE or FE members to solve bugs proactively. This points to high trust across functional teams.

3. **Trust in me as Backend Lead.** I believe the team trusts me, perhaps because of my work ethic and persistence in finding solutions, but I have not asked them. The fact that I review all backend code may also signal that I do not delegate enough.
4. **Contractors and EC staff.** We have no choice but to trust each other, and interests are aligned: the project's success matters to both, and EC staff who delegate poorly risk the outcome.
5. **Misunderstandings.** In four years I noticed only one or two misunderstandings, both minor setbacks.

Trust is high, though based on my observation rather than data. Part II can test it, for example by sharing code review with the two seniors.

### 1.3.3 Psychological Safety

Edmondson [9] defines psychological safety as a shared belief that the team is safe for interpersonal risk-taking. It combines with high standards to create a learning zone. I rate it medium to high, but I have not measured it.

1. **Evidence of safety.** People are encouraged to suggest better practices, share knowledge freely and ask for help. Leaders are open to feedback, and meetings focus on solving problems rather than defending positions.
2. **Contractor and employer risk.** Contractors depend on client recognition to stay, which may discourage them from challenging EC decisions. A team made up of several employers may also hide problems to protect its own firm.
3. **Incident handling.** Security work depends on early reporting of weaknesses. Incidents sometimes lead to blame, although a solution is always found and measures are taken to prevent a recurrence. The learning is good, but occasional blame can discourage early reporting.
4. **My code reviews.** I am the only reviewer of backend code, and the team has two senior developers, not juniors. They sometimes challenge my reviews, but not often enough. Even experienced developers may defer to my role, so my tone matters.
5. **Verdict and link to Part II.** Safety is medium to high, with two gaps: challenge of my reviews is too rare, and incidents sometimes lead to blame. Part II proposes a short pulse survey (Edmondson's seven items), blameless post-incident reviews and an explicit invitation to challenge my reviews.

### 1.3.4 Leadership Behaviours

Leaders on the project are versatile and open to feedback. I assess leadership through five behaviours.

1. **Openness to feedback.** When the Solutions Architect must make a major technical decision that affects my work, he almost always asks for my opinion, gives me the context and asks me to propose solutions. We then go back and forth until we agree on a final solution. This is leadership by setting context and sense-making, not by command and control.

2. **Decision-making.** The Tech Lead keeps the final decision, but it is reached through dialogue with the people affected. This helps people understand the reasons for decisions, although the back and forth can be slower than a direct decision.
3. **Support and recognition.** Leaders let the team focus on design instead of unproductive meetings, invite suggestions for new practices and appreciate good work. It is not clear that they also structure development and career paths.
4. **My own behaviour.** I combine advisory, coaching and hands-on roles. However, I review all backend code and do not delegate enough, and the two seniors may defer to my role. Whether my reviews coach or only gatekeep is a point for Part III.
5. **Verdict and link to Part II.** Leadership is medium to high. Openness is a clear strength, and my limited delegation is the main gap. Part II can address it by sharing code review with the two seniors.

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## 1.4 Summary of the Diagnosis

| Dimension | Readiness | Main evidence |
|---|---|---|
| Strategy | Medium | Strong purpose, but no measurable objectives or periodic review |
| Structure | Medium-High | Small, clear, cross-functional teams, but hierarchy of leads and review bottleneck |
| Systems | High | Modern DevOps toolchain, open source |
| Talent practices | Medium | Hybrid model gives flexibility, but turnover risk; strong retention |
| Culture | High | Modern, collaborative, innovation-oriented |
| Trust | High (unmeasured) | Peer help across teams, aligned interests, few misunderstandings |
| Psychological safety | Medium-High (unmeasured) | Encouragement to speak up, but contractor, incident and review risks unverified |
| Leadership behaviours | Medium-High | Open, collaborative decisions, but limited delegation by me |

Table 3: Summary of the agility readiness diagnosis (to refine)

**Link to Part II.** Gaps found so far are the implicit strategy, the code-review bottleneck, knowledge loss at contract end and the unknown depth of psychological safety. These will drive the interventions.

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